Symbolic Tumor-Growth ODE Candidate Ranking

Evaluation time: 2026-09-20 23:32 (UTC−05:00)

Basis: the attached evaluation instructions and randomized accepted-candidate CSV only. No refitting, trajectory simulation, hidden-data inference, or external sources were used.

Rank	Candidate ID	Main strength	Main concern	Reason for rank
1	C6	Simple single-log form with a moderate saturation scale and an additive offset.	MSE is higher than the two ultra-low-error complex fits.	Strong combination of simple structure, stable saturation behavior, and low training MSE without nested logs.
2	C4	Simple single-log form with a moderate offset and interpretable structure.	Slightly higher MSE than C6.	Nearly as accurate as C6 while retaining the same relatively robust, non-nested logarithmic structure.
3	C16	Very low training MSE and a smooth single-log-plus-linear form.	Large coefficients and substantial cancellation reduce numerical/parameter robustness and interpretability.	Its exceptional fit is important, but the coefficient magnitudes and cancellation are a robustness concern under the first criterion.
4	C21	Very low training MSE with a smooth log-based form and a stabilizing nonlinear factor.	Higher complexity and a comparatively large logarithmic scale parameter.	The fit is exceptionally close while the expression remains structurally smoother than nested-log candidates, but its complexity is substantial.
5	C24	Low MSE and a finite polynomial multiplier around a single logarithm.	Highest complexity tier and more parameter interactions.	Much better training agreement than most simple candidates, with fewer domain layers than nested-log forms, but limited parsimony.
6	C13	Best training MSE among the relatively moderate-complexity nested-log candidates.	Nested logarithms introduce additional domain sensitivity and complexity.	Its training agreement is strong, but the nested logarithmic structure is less robust than the leading single-log forms.
7	C19	Moderately low MSE with a polynomial-log structure.	More nonlinear interactions and higher complexity than the simple-log candidates.	The form can provide smooth saturation-like behavior, but its extra nonlinear structure weakens parsimony.
8	C22	Moderate MSE with a single logarithm multiplied by a low-order polynomial.	Complexity 19 and additional coefficient interaction.	Avoids nested logs and can represent changing growth strength, but the added polynomial factor is not strongly justified by the supplied MSE.
9	C14	Moderate complexity and a single logarithm with a linear volume term.	Training MSE is higher than C13 and the leading simple-log fits.	Its non-nested structure is relatively robust, but the added linear term gives less parsimony than C4/C6.
10	C11	Strong training MSE among nested-log candidates.	Nested logarithms add domain fragility and complexity.	Good data agreement, but the additional logarithmic layer makes the model less robust and interpretable than simpler alternatives.
11	C9	Good training MSE relative to most higher-complexity candidates.	Nested logarithms create additional domain restrictions.	Its fit is competitive, but the nested structure is a substantive robustness concern.
12	C7	Essentially the same structural behavior as a simple logarithmic saturation model.	Extra multiplicative parameter and higher complexity provide little training-MSE benefit over C4/C6.	The expression is structurally reasonable, but the extra parameter reduces parsimony without improving agreement materially.
13	C3	Very simple single-log structure with low complexity.	Training MSE is worse than C4/C6 and slightly worse than C2.	Strong parsimony and non-nested domain behavior keep it competitive despite its higher error.
14	C2	Very simple single-log structure with low complexity.	Training MSE is higher than C3/C4/C6.	Its robustness and parsimony are clear strengths, but the supplied data favor several nearby alternatives.
15	C1	Very simple single-log structure with low complexity.	Highest MSE among C1-C4.	Highly parsimonious and non-nested, but its training agreement is weaker than the other simple-log candidates.
16	C12	Single outer logarithm with an inner logarithmic argument shifted by a constant.	Nested logarithmic domain sensitivity and higher complexity.	The additional shift gives flexibility, but the supplied MSE does not compensate for the added structural fragility.
17	C17	Triple-nested logarithmic structure with an additive constant.	Triple nesting and complexity 17 increase domain sensitivity.	It is accepted and has moderate MSE, but the structure is considerably less robust and interpretable.
18	C23	Polynomial multiplier with a single logarithm and moderate MSE.	High complexity and more coefficient interactions.	Its non-nested logarithm is preferable to nested logs, but the training error and complexity do not support placing it higher.
19	C10	Triple/nested logarithmic structure with the same general family as C9/C11.	High MSE plus nested-log domain sensitivity.	Both accuracy and structural simplicity are weaker than the stronger nested-log candidates.
20	C20	Polynomial-log structure similar to C19.	Higher MSE than C19 with the same complexity.	The extra nonlinear structure is not supported by the supplied training MSE relative to C19.
21	C5	Simple single-log-plus-offset form.	High training MSE despite moderate complexity.	Its non-nested form is relatively robust, but the supplied fit is substantially worse than nearby alternatives.

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22	C18	Triple-nested logarithmic form with an additive constant.	High MSE, triple nesting, and complexity 17.	It combines substantial structural fragility with weak training agreement relative to the better nested-log candidates.
23	C8	Single-log form with a fitted multiplicative coefficient.	Highest MSE among the complexity-11 pair.	Its structure is not intrinsically as fragile as nested logs, but its poor training agreement makes it less defensible.
24	C15	Linear-plus-log structure with moderate complexity.	Very high training MSE.	The non-nested structure is a positive feature, but the supplied training agreement is too weak to offset its added complexity.

Top choice: C6. C6 combines a relatively simple, non-nested logarithmic form with a low training-trajectory MSE, while avoiding the extra domain sensitivity and parameter interactions introduced by the more complex candidates. Its main limitation is that C16 and C21 have dramatically smaller supplied training MSE values, but their larger coefficient/structural complexity raises robustness and interpretability concerns under the stated first criterion.

The candidate set spans simple single-log models, nested-log models, and more nonlinear polynomial-log forms. The strongest training fits are concentrated in the higher-complexity expressions, whereas several lower-complexity candidates offer simpler non-nested structures with somewhat larger MSE; the ranking therefore reflects the requested ordering of robustness, training agreement, and parsimony rather than MSE alone.